Prediction of a Stream Water Level using Meteorological Data

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1 Context and Scope of the Project

The "Bayerische Akademie der Wissenschaften" maintains since 1973 a high alpine environmental observatory around the "Vernagtferner", a glacier located in the Ötztal Alps in Tyrol, Austria. The "Vernagtferner" nourishes the "Vernagtbach" and the measurement station "Pegelstation Hydrologie" measures the water level of this stream. The closeby measurement station "Pegelstation Meteorologie" measures meteorological variables related to air temperature, atmospheric pressure, humidity, precipitation, snow height, wind and radiation. Both stations are in 2640 m. There is another measurement station called "Schwarzkögele" in a height of 3080 m in a distance of about 1 km from the "Pegelstation Hydrologie" and "Pegelstation Meteorologie". There also exist measurement stations on the ice of the glacier, but because of the harsh weather conditions they cannot always deliver continuous data.

The objective of this data science project is to analyze the feasibility of predicting the water level of the "Vernagtbach" measured by "Pegelstation Hydrologie" from the available meteorological data measured by "Pegelstation Meteorologie" and "Schwarzkögele". This could be useful to be able to get information about the water level of the "Vernagtbach" without actually measuring it.

1.1 Special Characteristics of Hydrological Systems

Hydrological systems exhibit several important characteristics:

- Temporal dependencies
- Delayed responses to rainfall and snow melt
- Seasonal effects
- Nonlinear relationships

For example, precipitation does not immediately affect the stream level. Instead, the response may occur several hours later due to infiltration, runoff processes, and storage effects in the catchment.

Therefore, past observations must be incorporated into the predictive model.

2 Data Description

The available data comprises measurements every 5 minutes for 12 years from January 1, 2013 until December 31, 2024 for the measurement stations

"Pegelstation Hydrologie" and "Pegelstation Meteorologie" and for over 9 years from July 22, 2015 until December 31, 2024 for the measurement station "Schwarzkögele".

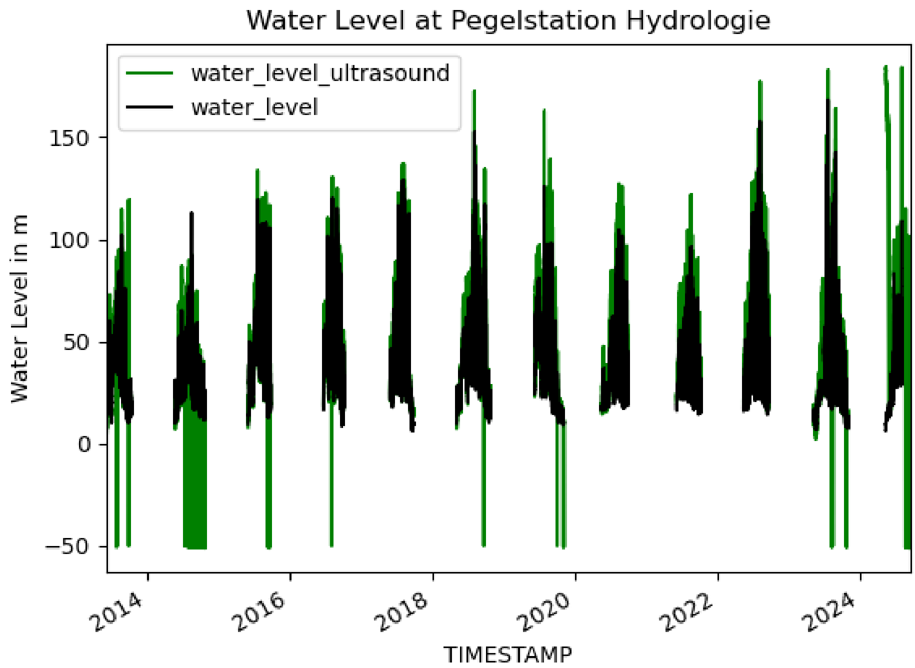
The raw data includes the following measurement data from the 3 measurement stations "Pegelstation Hydrologie", "Pegelstation Meteorologie" and "Schwarzkögele" and is exclusively numeric:

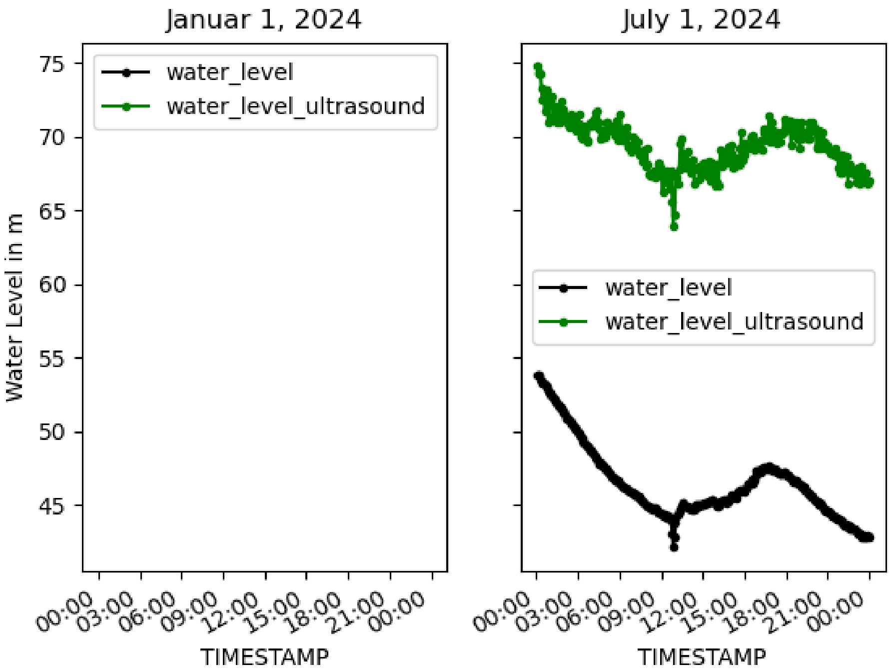
- "Pegelstation Hydrologie":
  - water level
from a gauge measurement (water_level)
  - water level
from a measurement with ultrasound (water_level_ultrasound)
- "Pegelstation Meteorologie":
  - air temperature (PM_temperature)
  - atmospheric pressure (PM_atmospheric_pressure)
  - relative humidity (PM_relative_humidity)
  - precipitation (PM_precipitation)
  - cumulative precipitation (PM_precipitation_cum)
  - snow height (PM_snow_height)
  - wind speed (PM_wind_speed)
  - wind direction (PM_wind_direction)
  - short wave radiation downwards (PM_SWD)
  - short wave radiation upwards (PM_SWU)
- "Schwarzkögele"
  - air temperature (SK_temperature)
  - relative humidity (SK_relative_humidity)
  - wind speed (SK_wind_speed)
  - wind direction (SK_wind_direction)

2.1 Presentation of the Raw Data

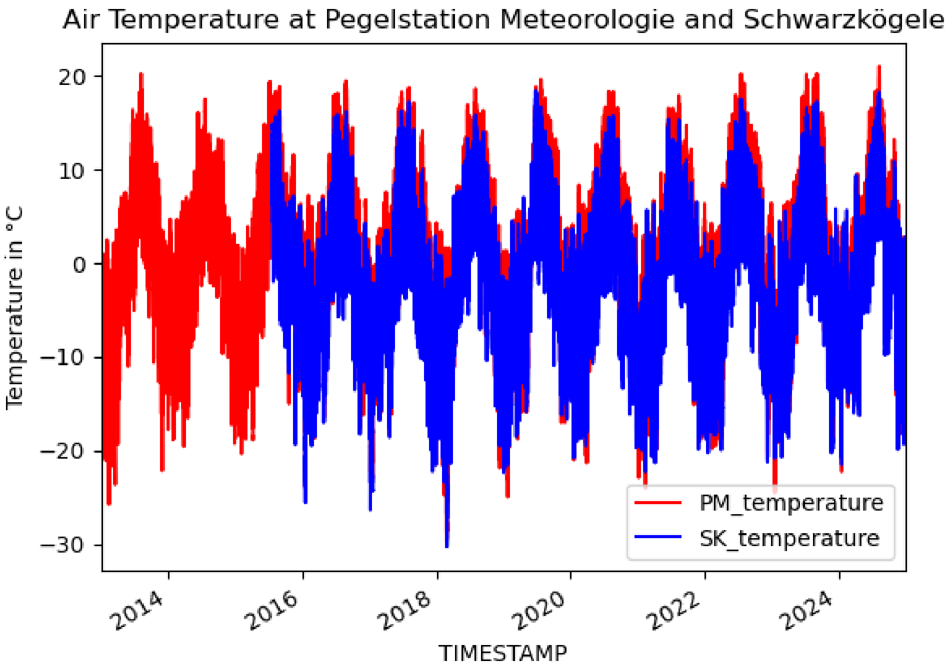
The raw data is presented with 3 plots for corresponding variables. One plot shows the data for the whole time period. The other two plots show exemplarily the values for January 1, 2024 and July 1, 2024.

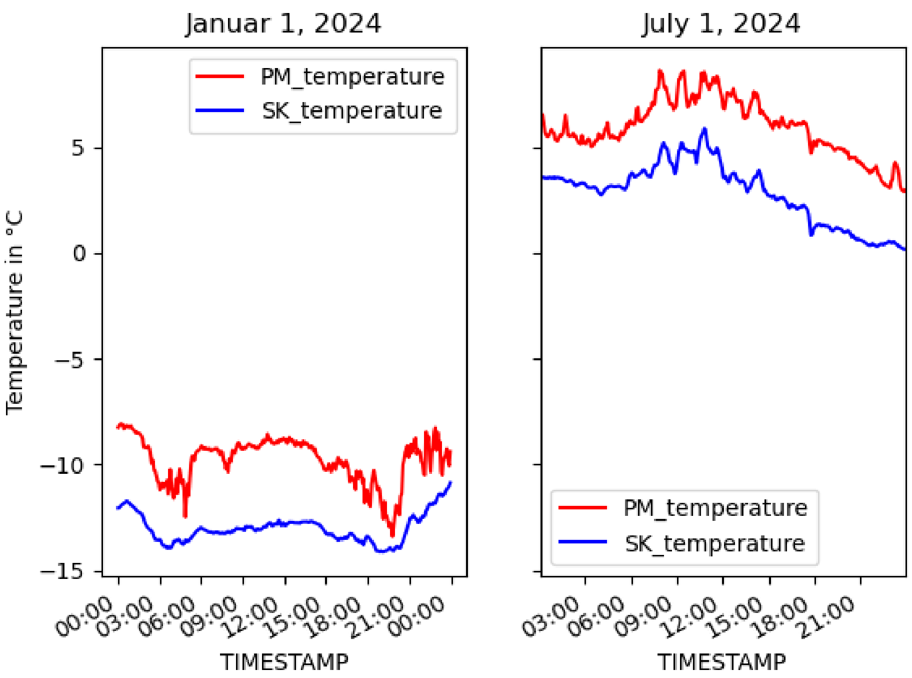
Water Level There are no measurements of the water level in winter as the measurement instruments do not work when the air temperature is below zero and snow has fallen. For these times a low constant water level can be safely assumed. The measurement of the level with ultrasound still contains faulty values. There is a pronounced offset between the two measurement methods. The values of water_level_ultrasound are to be used to fill the gaps with missing values of water_level.



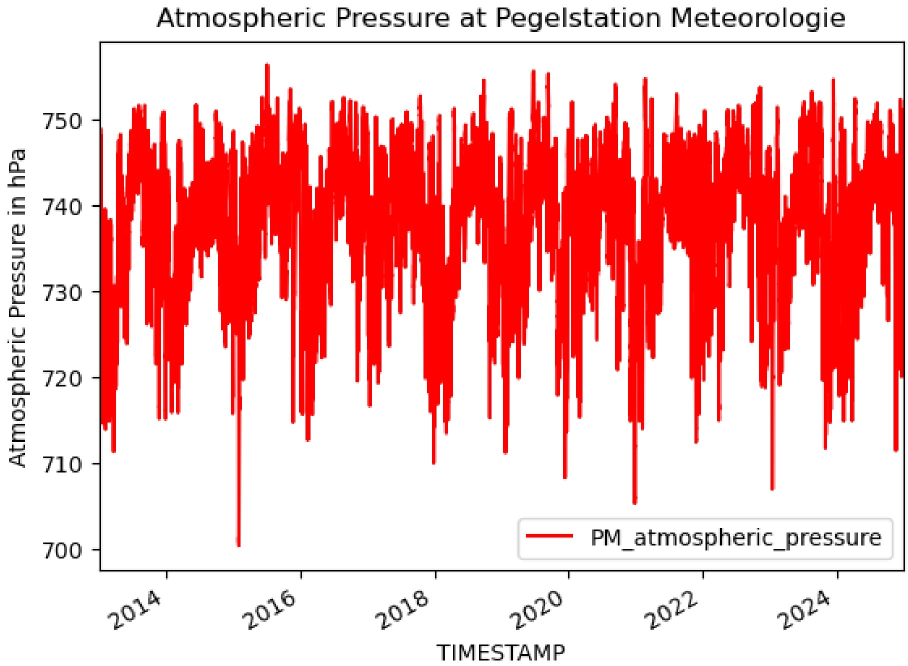


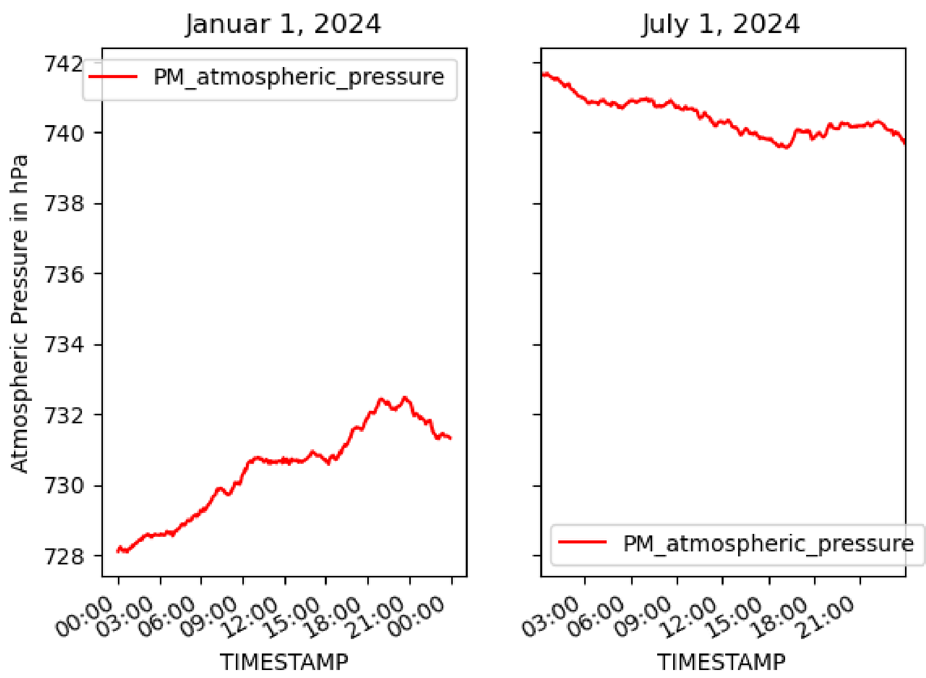
Air Temperatures The air temperature of Schwarzkögele is not available in the first 2 years and generally lower than the air temperatures at Pegelstation Meteorologie. This is due to the greater height of Schwarzkögele.


Air Temperature at Pegelstation Meteorologie and Schwarzkögele
Temperature in °C
20
10
0
−10
−20
−30
2014
2016
2018
2020
2022
2024
TIMESTAMP
PM_temperature
SK_temperature

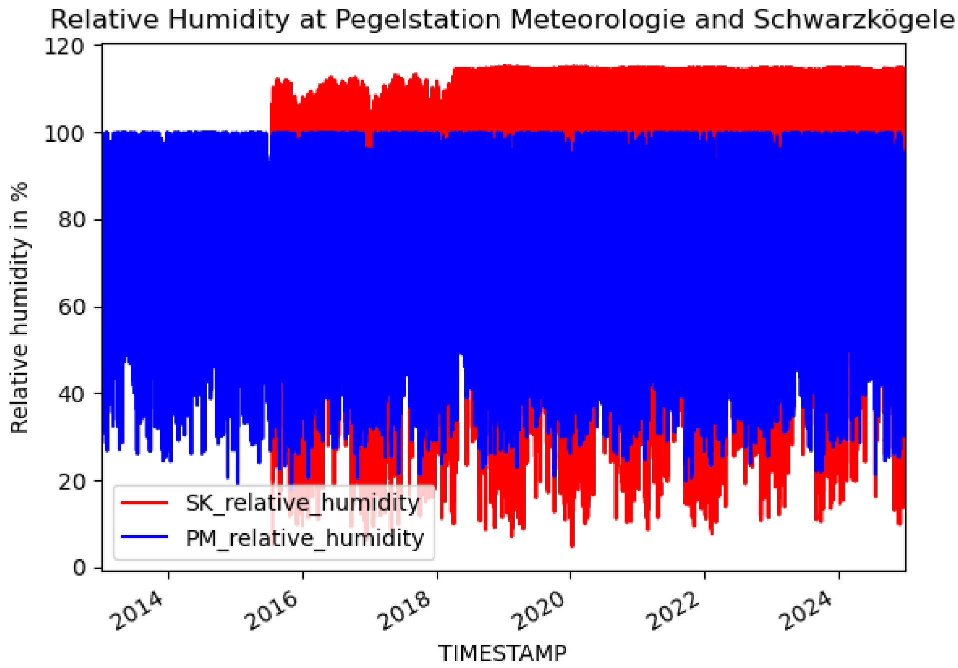

Januar 1, 2024
July 1, 2024
Temperature in °C
5
0
−5
−10
−15
00:00
03:00
06:00
09:00
12:00
15:00
18:00
21:00
00:00
TIMESTAMP
PM_temperature
SK_temperature
03:00
06:00
09:00
12:00
15:00
18:00
21:00
TIMESTAMP
PM_temperature
SK_temperature

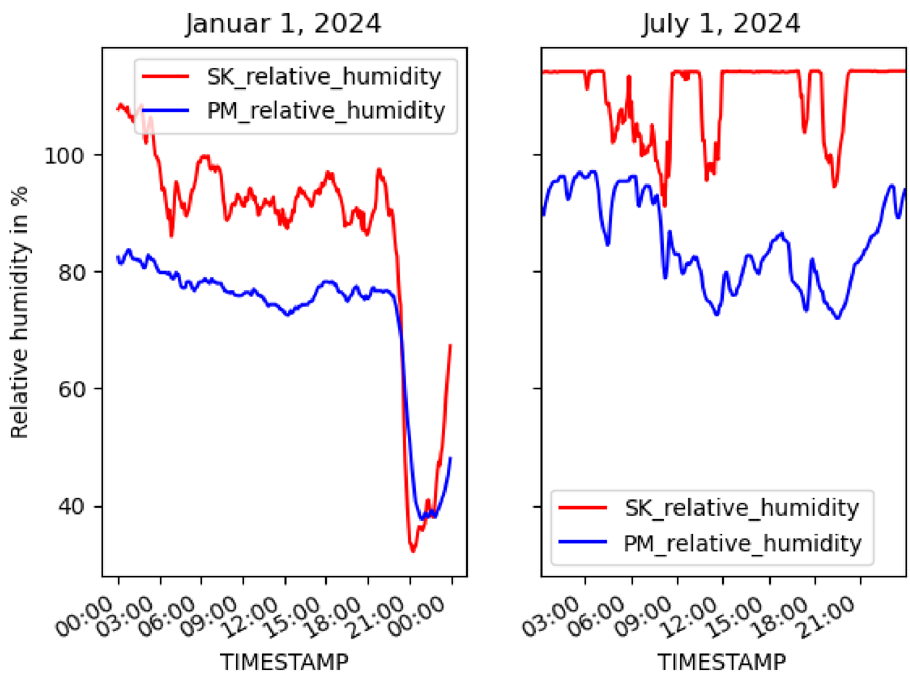
Atmospheric Pressure



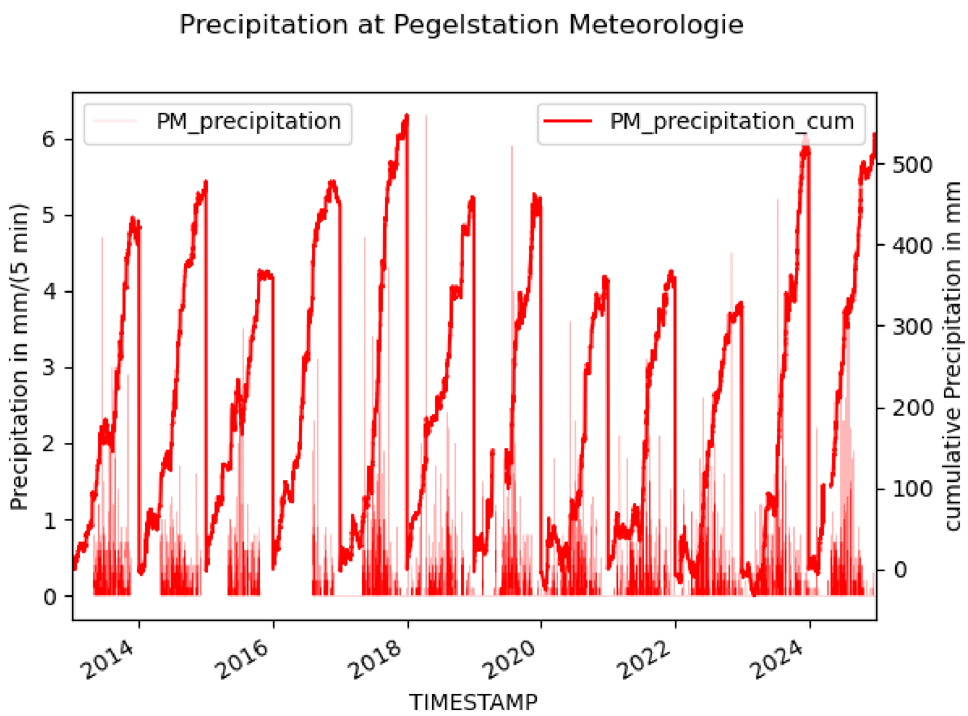


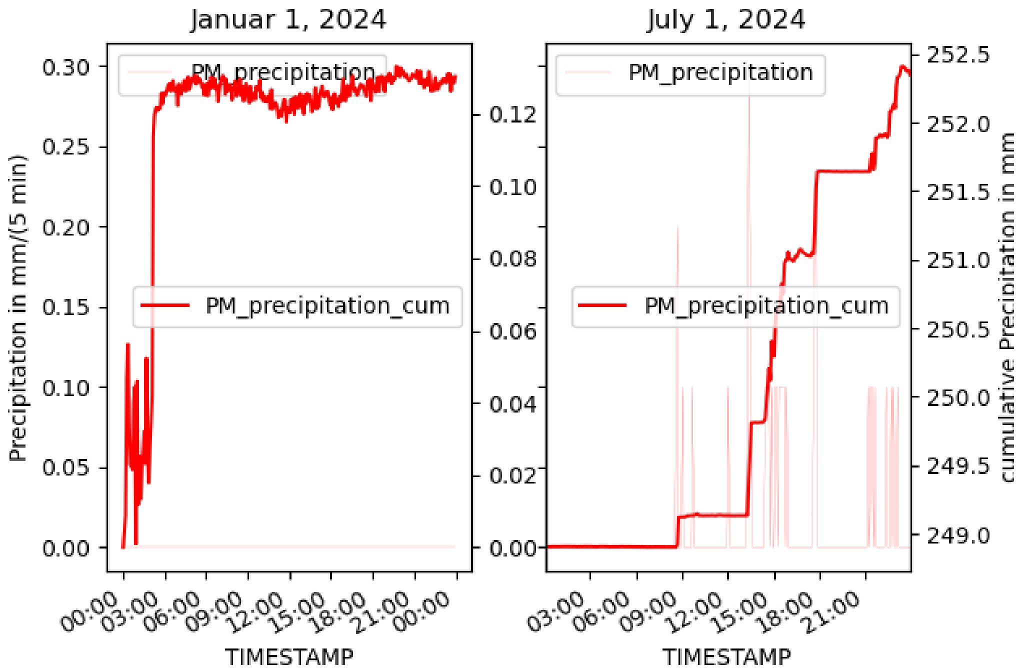
Relative Humidity The plots for relative humidity show a great difference between the measurement stations Schwarzkögele and Pegelstation Meteorologie. As relative humidity should be in the range between 0 and 100 and SK_relative_humidity shows a different behavior from some time in 2018 SK_relative_humidity has to be regarded invalid and should not be used for modeling.


Relative Humidity at Pegelstation Meteorologie and Schwarzkögele
Relative humidity in %
SK_relative_humidity
PM_relative_humidity
TIMESTAMP

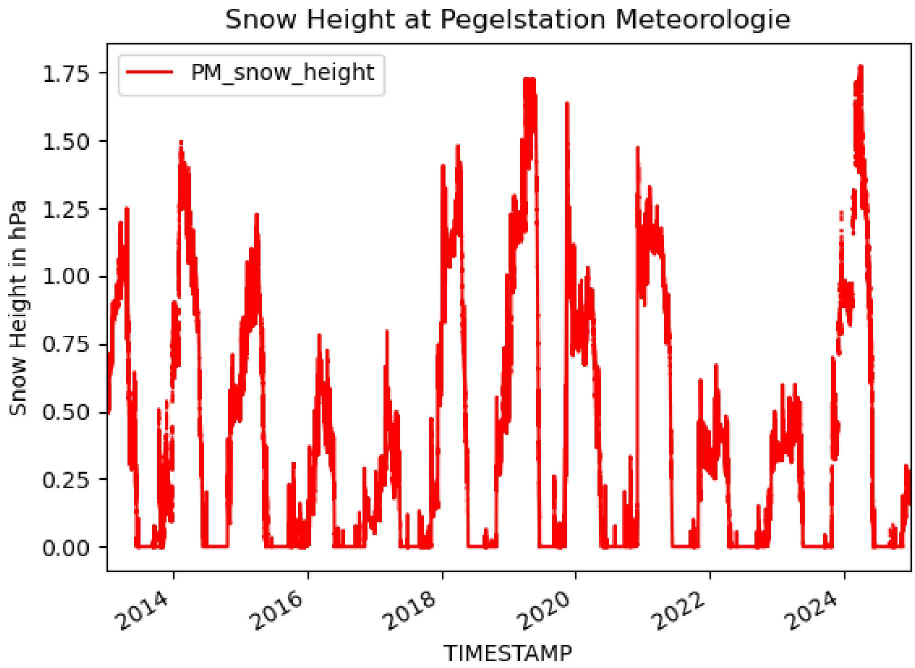

Januar 1, 2024
July 1, 2024
Relative humidity in %
SK_relative_humidity
PM_relative_humidity
TIMESTAMP

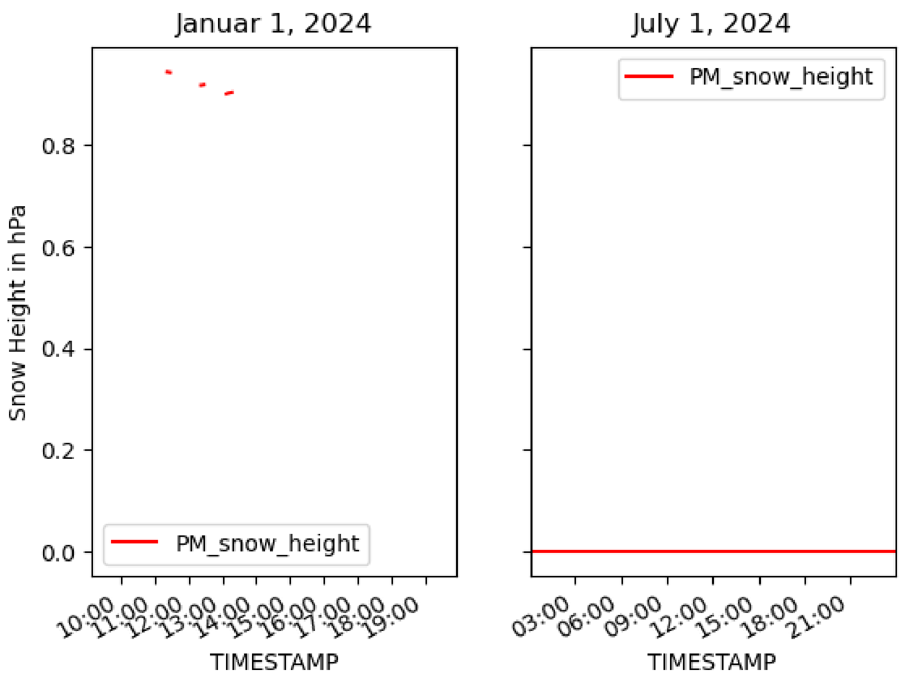
Precipitation The precipitation PM_precipitation measured in mm/(5 min) has measurement gaps during the winter in the earlier years because the measurement instrument was then disassembled in winter. The precipitation PM_precipitation_cum measured in mm is the cumulative precipitation during a year and is corrupted by the evaporation of water from the measurement instrument. PM_precipitation_cum shows a significant measurement gap in 2019. As the precipitation is measured in two ways it should be sufficient to keep one measurement.



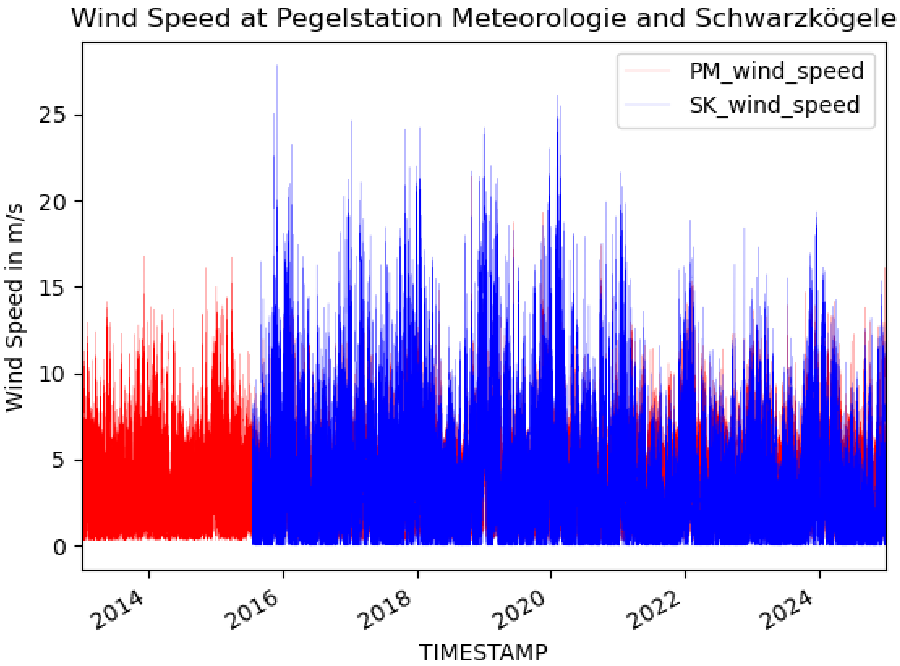


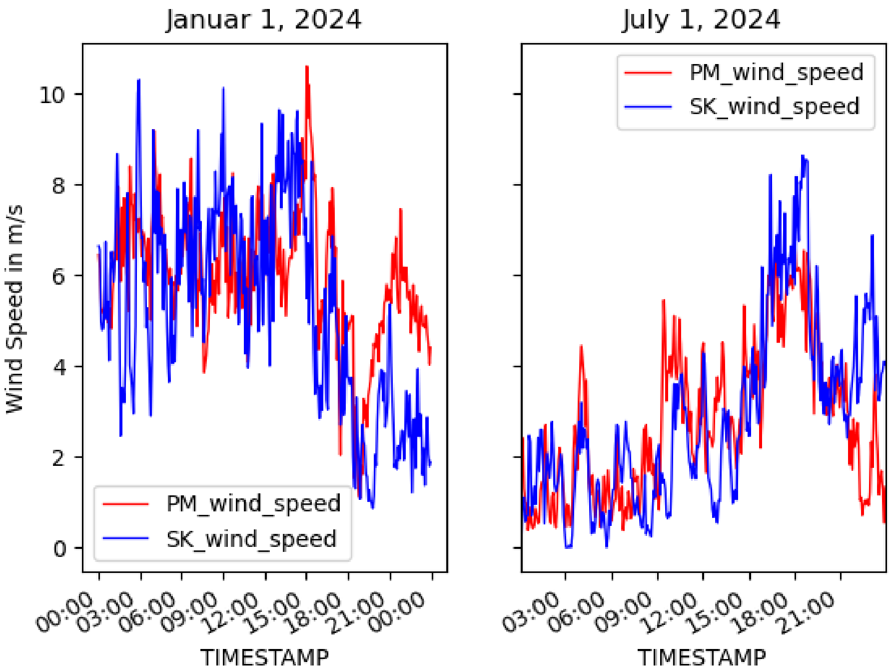
Snow Height The snow height exhibits several missing values. However as the snow height does not show larger variations over time it should be safe to interpolate missing values.


Snow Height at Pegelstation Meteorologie
PM_snow_height
Snow Height in hPa
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2014
2016
2018
2020
2022
2024
TIMESTAMP

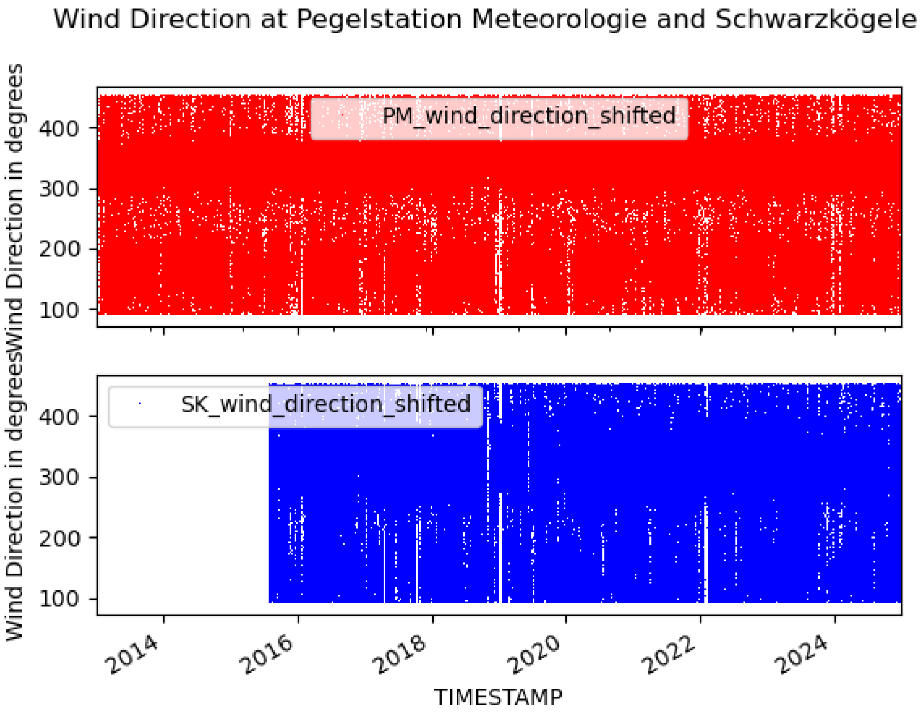

Januar 1, 2024
July 1, 2024
PM_snow_height
PM_snow_height
Snow Height in hPa
0.8
0.6
0.4
0.2
0.0
10:00
11:00
12:00
13:00
14:00
15:00
16:00
17:00
18:00
19:00
03:00
06:00
09:00
12:00
15:00
18:00
21:00
TIMESTAMP
TIMESTAMP

Wind Speed The wind speeds measured at Pegelstation Meteorologie and Schwarzkögele show a little offset probably because of the use of different measurement instruments.

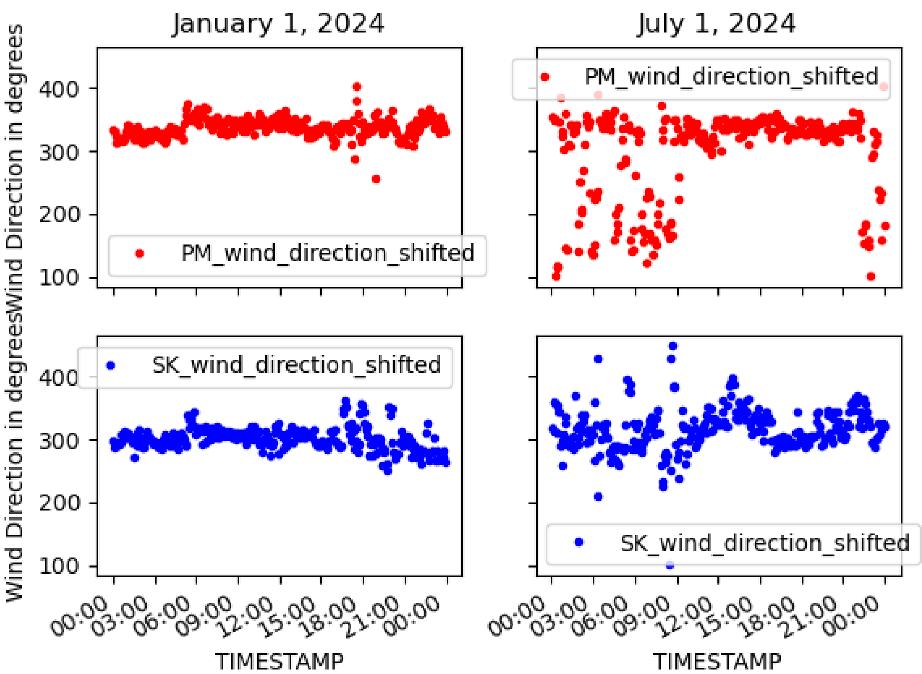




Wind Direction There are 2 main wind directions around 150 and 330 degrees at the measurement stations which are more pronounced at Pegelstation Meteorologie located deeper down the glacier valley.

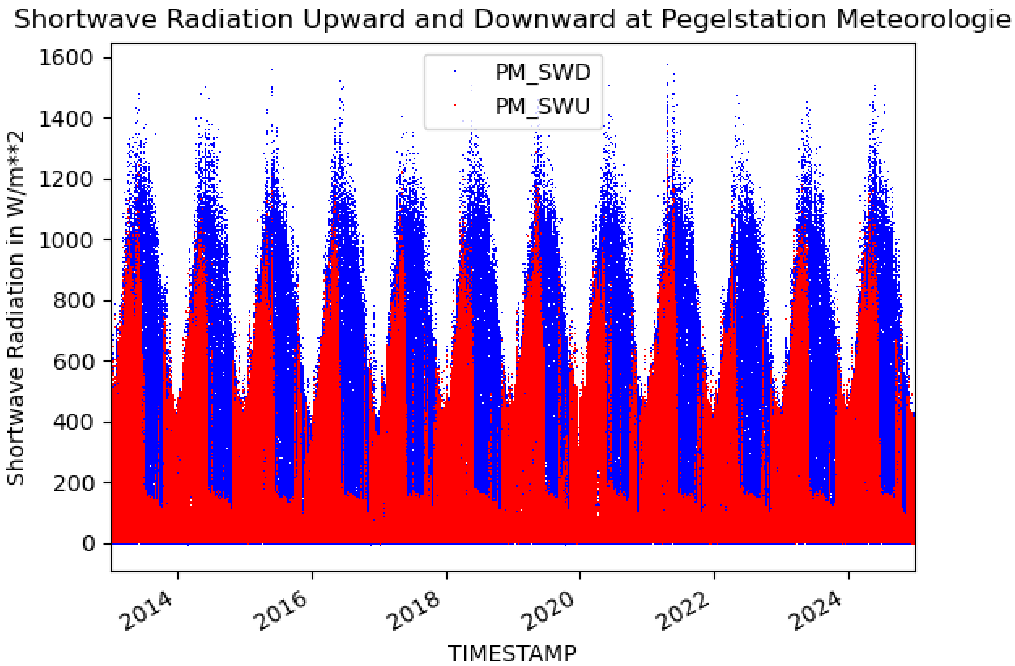


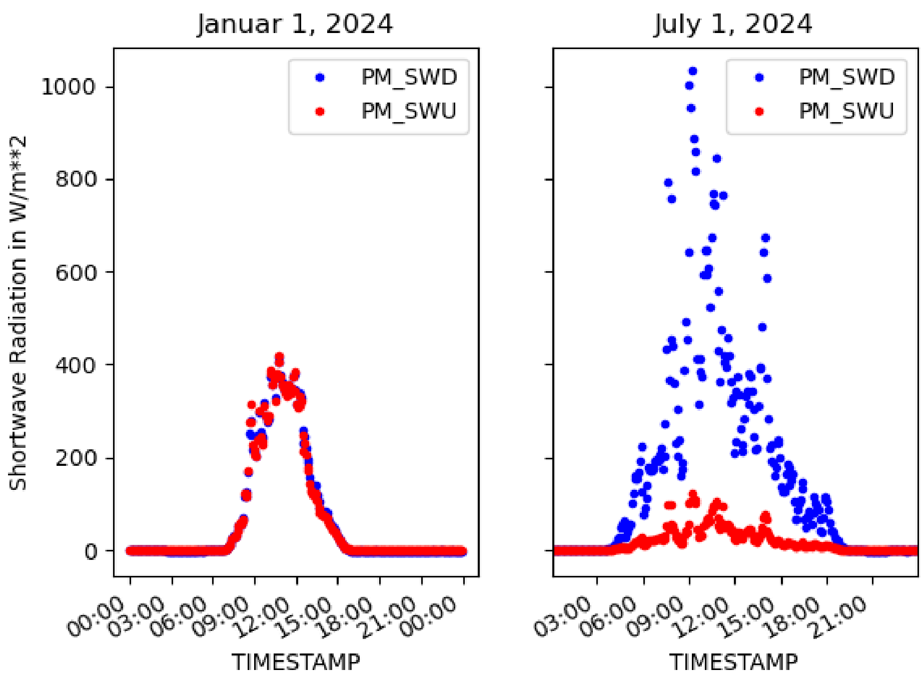
Wind Direction at Pegelstation Meteorologie and Schwarzkögele
PM_wind_direction_shifted
SK_wind_direction_shifted
Wind Direction in degrees
400
300
200
100
2014
2016
2018
2020
2022
2024
TIMESTAMP



January 1, 2024
July 1, 2024
PM_wind_direction_shifted
SK_wind_direction_shifted
Wind Direction in degrees
400
300
200
100
00:00
03:00
06:00
09:00
12:00
15:00
18:00
21:00
00:00
TIMESTAMP

Short Wave Radiation





3 Data Preprocessing

3.1 Preprocessing of the water level (Target Variable)

For short periods of missing values the missing values of water_level are interpolated. Longer gaps of missing values of water_level are filled by the values of water_level_ultrasound that are adjusted to fit the last and first values of non missing values of the water_level.

3.2 Timestamp Handling

Convert timestamps to datetime format and sort chronologically.

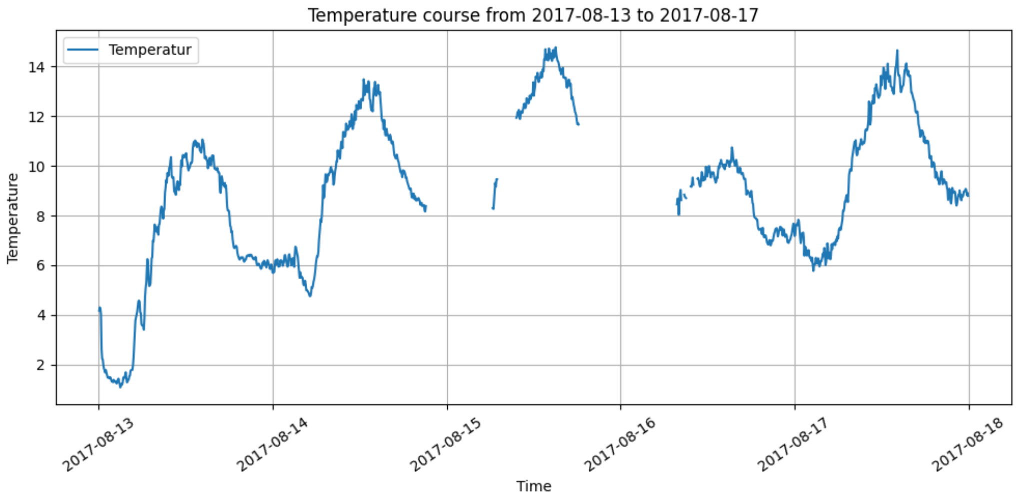
3.3 Missing Values

For the describtive variables the two methods:

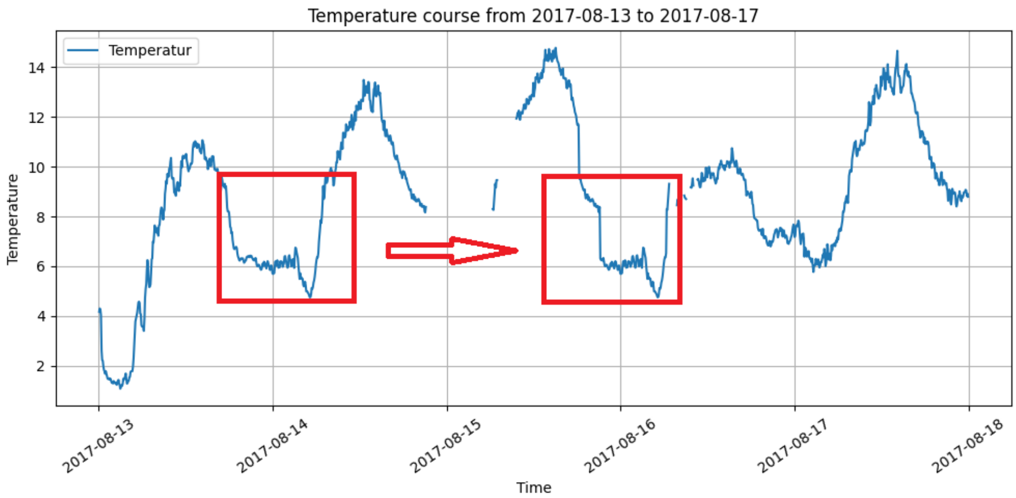
- Forward shifting
- Forward filling

Were applied.

For Temperature, Atmospheric pressure, Humidity, Wind speed, Wind direction and Solar radiation, missing values, which time period was within O(one day), were filled by data from one or two days before the data gap (Forward shifting). The assumption is, that the variables do not change significantly in the considered time interval. This approach is describe in the following pictures.

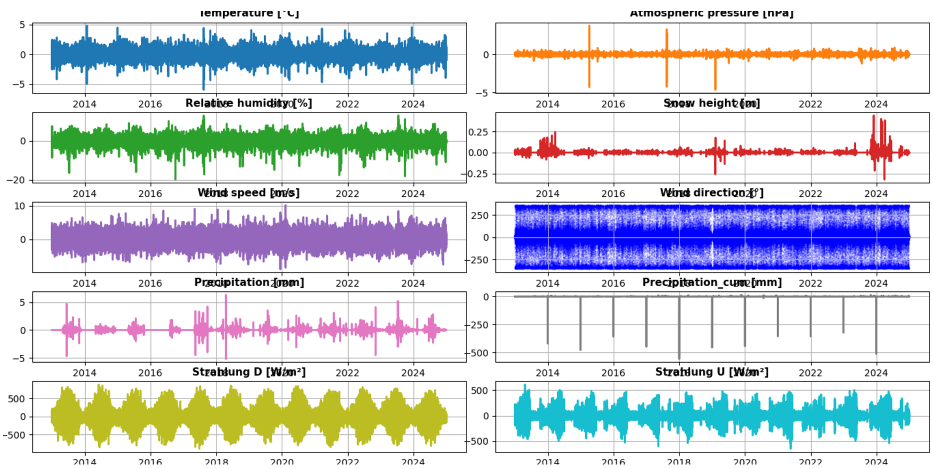


The picture shows the raw data distribution of the temperature in the time interval 2017-08-13 to 2017-08-17 with gaps in the temperature course. The biggest gap is from 2017-08-15 18:17:30+00:00 to 2017-08-16 06:42:30+00:00. Which was determined by a python script counting nan to find the biggest gaps.



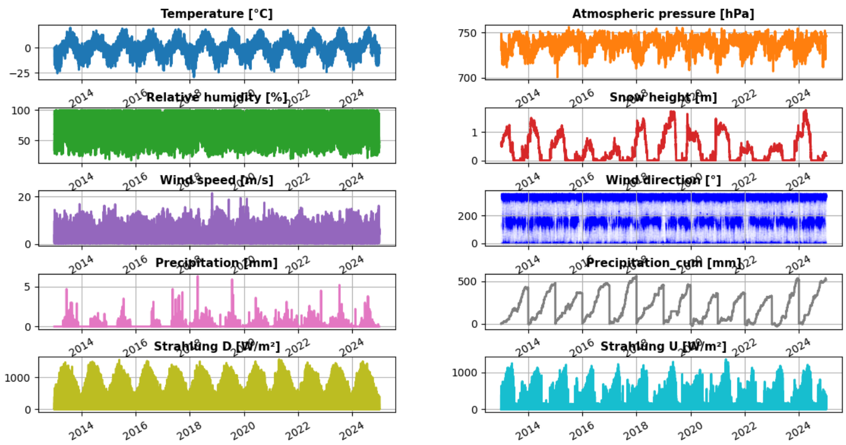
Like the above figure shows leads this method to discontinuities at the points where the data is connected (current / history data). With the above assumption the expectation is, that this jump is in a tolerable range (e.g. dt¡5k, dp¡1hPa). For the above example the jump of the temperatue is 0.9°C over 2h.

The variation in the differences between consecutive measurement values should remain within the range of the raw (unmodified) data. The expected value of the measurement differences should be zero within the limits of the measurement chains' accuracy.

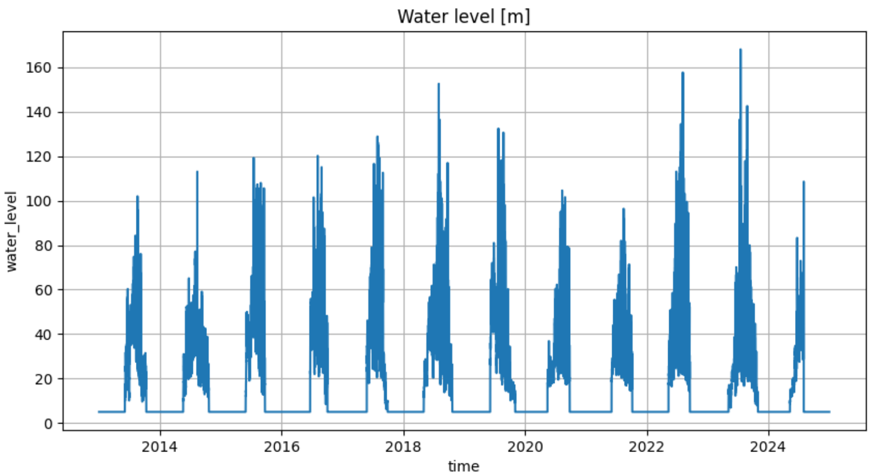


The variation in the values is due to physical factors, such as actual temperature fluctuations at the measurement site. A time-stable fluctuation including seasonal effects should be observable over the measurement time period. The magnitude of the variations depend on the specific measurement variable, see figure above.

The remaining gaps were filled by forward filling. Final result see picture below.



The preprocessed water level is shown in the next picture.



3.4 Data Quality Checks

Perform plausibility checks:

- Negative precipitation values
- Unrealistic sensor jumps

This issues are not detected, see table xy.

4 Exploratory Data Analysis